

第四章. 12 依题意, 设每次需要调整设备概率为 p 备注: 关于正态分布.

$$p = 1 - 0.9^{10} = 1 - 10 \times 0.9^9 \times 0.1 = 1 - 0.349 = 0.651$$

$$X \sim B(4, p) \quad E(X) = np = 1.056$$

$$\begin{array}{r} 0.349 \\ + 0.387 \\ \hline 0.736 \end{array} \quad \begin{array}{r} 0.264 \\ \times 2.1 \\ \hline 1.056 \end{array}$$

备注: 关于二项分布, 期望的计算

$$公式 E(X) = np$$

$$E(X) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$X = \begin{cases} 1 & \text{第 } i \text{ 次试验成功} \\ 0 & \text{第 } i \text{ 次试验失败} \end{cases}$$

$$X = X_1 + X_2 + \dots + X_n$$

$$P\{X_i = 1\} = p \quad P\{X_i = 0\} = 1 - p \quad \text{补: 列出 } E(X)$$

$$E(X) = \sum_{i=1}^n E(X_i) = np$$

$$P_k = (1-p)^4 \cdot 4p(1-p)^3 \cdot 4p^3(1-p) \cdot p^4$$

$$5. f(x) = \begin{cases} \frac{1}{1500}x, & 0 \leq x \leq 1500 \\ -\frac{1}{1500}(x-3000), & 1500 < x \leq 3000 \\ 0, & \text{其它} \end{cases}$$

$$E(X) = \int_0^{\infty} x f(x) dx$$

$$E(X) = \int_0^{1500} x f(x) dx + \int_{1500}^{3000} x f(x) dx$$

$$= \int_0^{1500} \frac{1}{1500} x^2 dx + \int_{1500}^{3000} -\frac{1}{1500} x(x-3000) dx$$

$$= \frac{1}{1500} \left[\frac{1}{3} x^3 \right]_0^{1500} + \left[-\frac{1}{3000} x^3 + x^2 \right]_{1500}^{3000}$$

$$= \frac{1}{1500} \left(\frac{1}{3} \times 1500^3 \right) + \left(-\frac{1}{3000} \times 3000^3 + 3000^2 \right) - \left(-\frac{1}{3000} \times 1500^3 + 1500^2 \right)$$

$$= \frac{1}{1500} \left[2 \times 1500^2 \times 1500 - 3000^2 + \frac{1}{2} \times 1500^2 + 3000 \times 1500 \right]$$

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X	-2	0	2
P _k	0.4	0.3	0.3

$$E(X) = \sum_{k=1}^3 x P_k = -2 \times 0.4 + 2 \times 0.3 = -0.2$$

$$E(X^2) = \sum_{k=1}^3 x^2 P_k = 4 \times 0.4 + 0.3 = 2.8$$

$$E(3X^2 + 5) = 5 \times 0.3 + (3 \times 4 \times 0.7) = 11.5$$

$$= 11.5 + 11.9 = 23.4$$

$$E(X) = 0 \quad D(X) = 9$$

$$E(X^2) = D(X) + [E(X)]^2 = 9$$

$$E(Y) = E(5X^2) = 5 E(X^2) = 22.5$$

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$$E(AX+B) = AE(X) + B$$

$$Z = \frac{X-\mu}{\sigma} \quad \varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$$

$$E(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}} dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} d\left(-\frac{t^2}{2}\right) = 0$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^u du = 0$$

$$E(X) = E(\sigma Z + \mu) = \sigma E(Z) + \mu = \mu$$

$$D(Z) = E(Z^2) - [E(Z)]^2$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-\frac{t^2}{2}} dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}} d\left(-\frac{t^2}{2}\right) = 1$$

$$= \frac{1}{\sqrt{2\pi}} \left[t e^{-\frac{t^2}{2}} \right]_{-\infty}^{\infty} + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt$$

$$= 1$$

$$D(X) = D(\sigma Z + \mu) = \sigma^2$$

$$= E(X^2) - [E(X)]^2$$

$$14. (1) E(X_1) = \int_0^{\infty} x f_1(x) dx$$

$$= \int_0^{\infty} 2x e^{-2x} dx$$

$$= \int_0^{\infty} -x de^{-2x}$$

$$= [-x e^{-2x}]_0^{\infty} - \int_0^{\infty} e^{-2x} d(-x)$$

$$= \int_0^{\infty} e^{-2x} dx$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t} dt$$

$$= \frac{1}{2} \left[-e^{-t} \right]_0^{\infty} = \frac{1}{2}$$

$$E(X_2) = \int_0^{\infty} x f_2(x) dx = \int_0^{\infty} 4x e^{-4x} dx$$

$$= \int_0^{\infty} e^{-4x} dx$$

$$= \frac{1}{4} \int_0^{\infty} e^{-t} dt = \frac{1}{4}$$

$$E(X_1 + X_2) = E(X_1) + E(X_2) = \frac{3}{4}$$

$$E(X_2^2) = \int_0^{\infty} x^2 f_2(x) dx$$

$$= \int_0^{\infty} x^2 e^{-4x} dx$$

$$= \int_0^{\infty} -x^2 de^{-4x} = \left[-x^2 e^{-4x} \right]_0^{\infty} + \int_0^{\infty} e^{-4x} 2x dx$$

$$= 0 + \frac{1}{2} \int_0^{\infty} 4x e^{-4x} dx$$

$$= \frac{1}{2} \times E(X_2) = \frac{1}{8}$$

$$E(2X_1 - 3X_2^2) = 2E(X_1) - 3E(X_2^2)$$

$$= 1 - 3 \times \frac{1}{8} = \frac{5}{8}$$



24. $E(\sum_{i=1}^n X_i) = 30n$ $D(\sum_{i=1}^n X_i) = 2.5^2 n$

总质量 $Y = \sum_{i=1}^n X_i$

$Y \sim (50n, 2.5^2 n)$

(2) 又 X_1, X_2 相互独立
 $E(X_1 X_2) = E(X_1) E(X_2)$
 $= \frac{1}{8}$

22. (1) $E(Y) = 2E(X_1) - E(X_2) + 3E(X_3) - \frac{1}{2}E(X_4)$
 $= 2 - 2 + 9 - 2 = 7$

方差都是不断在相加的。
 $D(Y) = 4D(X_1) - D(X_2) + 9D(X_3) + \frac{1}{4}D(X_4)$
 $= 16 - 3 + 9 \times 2 + \frac{1}{4}$
 $= 31 + \frac{1}{4} = \frac{125}{4} = 31.25$

24. (2) $Z = \frac{X - \mu}{\sigma} = \frac{X - 720}{30}$

$X = \mu + \sigma Z_1 = 720$

$E(X) = 720$ $E(Y) = 640$

$D(X) = 30^2$ $D(Y) = 25^2$

$Z_1 = 2X + Y$

$E(Z_1) = 720 \times 2 + 640 = 1440 + 640 = 2080$

$Z_2 = X - Y$

$E(Z_2) = 720 - 640 = 80$

$D(Z_1) = 4D(X) + D(Y)$ $D(Z_2) = D(X) + D(Y)$ 29. 验证不相关

$= 4 \times 900 + 625$

$= 3600 + 625$

$= 4225$

$= 900 + 625$

$= 1525$

$D(X+Y) = D(X) + D(Y) + 2E\{(X-E(X))(Y-E(Y))\}$

$COV(X, Y) = E\{(X-E(X))(Y-E(Y))\}$

备注: $X_1, \dots, X_n$ 相互独立, 且则有 $D(X_1 \pm \dots \pm X_n) = D(X_1) + D(X_2) + \dots + D(X_n)$

$= D(X_1) + D(X_2) + \dots + D(X_n)$ $Z_2 \sim (80, 1525)$ $E(X) = -1 \times \frac{3}{8} + 1 \times \frac{3}{8} + 0 = 0$

$P: Z_1 \sim (2080, 4225)$ $P\{X > Y\} \sim (80, 1525)$

$COV(X, Y) = E\{XY\}$

$= 1 \times \frac{2}{8} + 0 \times \frac{4}{8} + -1 \times \frac{2}{8} = 0$

故不相关。不独立。易见 $P\{X=0, Y=0\} = 0$

$D(X-E(XY)) \neq P\{X=0\}P\{Y=0\}$ 故 X, Y 不是

$P(X, Y) \cdot F(X, Y) \Rightarrow F(0, 0) = 0$ 相互独立的。

$F_X(0) = \frac{1}{4}$ $F_Y(0) = \frac{1}{4}$ 举一个反例

$F(0, 0) \neq F_X(0)F_Y(0)$

故不是相互独立的

X, Y 独立 $\Rightarrow A, B$ 独立

$Z_2 = X - Y = P\{X - Y > 0\}$

$\frac{0-80}{\sqrt{1525}} = -0.8$
 $P\{X > Y\} \approx \Phi(1-0.8) = 1 - \Phi(0.8) = \frac{1}{\sqrt{2\pi}} \int_{0.8}^{\infty} e^{-\frac{t^2}{2}} dt$
 $= \Phi(2.0548) = 0.9798$

$E(Z_3) = 720 + 440 = 1160$

$Z_3 \sim (1160, 5 \times 720^2)$ (2.048)

$P\{X+Y > 1400\} = 1 - \Phi(1.025) = 1 - 0.8461$

$\frac{1400-1160}{39} = 1.025$

30. $P_{XY} = \frac{COV(X, Y)}{\sqrt{D(X)} \sqrt{D(Y)}}$

$D(X) = 0.5 = D(Y)$

$P_{XY} = 0$ $COV(X, Y) = 0$

$E(X) = P(A)$ $E(X)E(Y)$

$E(Y) = P(B)$ //

$E(XY) = P(A)P(B)$

$= P(A)P(B)$

$E((X-P(A))(Y-P(B)))$

$= E(XY) - P(B)E(X) - P(A)E(Y) + P(A)P(B) = 0$

